Distribution Assignment

Q1. Simulate 30 rolls with = RANDBETWEEN (1,6). What is the probability of rolling a 3 exactly 5 times? (Hint: Use BINOM.DIST)

Answer – Check the Attached Excel Sheet for the Practical Solution.

Q2. Generate 100 values in Excel using the continuous uniform distribution RAND() and plot a histogram. Describe the shape of the distribution.

Answer – For the Histogram check attached Excel Sheet
The Shape of the Distribution looks approximately Uniform

Q3. A dataset has a mean of 50 and a standard deviation of 5. What percentage of values lie between 45 and 55 if the data follows a normal distribution?

Answer – This one uses the Empirical Rule for a Normal Distribution.

Step 1: Identify how far the range is from the mean

- Mean = 50
- Standard deviation = 5

Range given: 45 to 55

Calculate distance from the mean:

- $50-5=45$
- $50+5=55$

So the interval 45–55 = mean ± 1 standard deviation.

Step 2: Apply the Empirical Rule

For a normal distribution:

- 68% of values lie within ±1 standard deviation
- 95% within ±2 standard deviations
- 99.7% within ±3 standard deviations

Final Answer

About 68% of the values lie between 45 and 55.

Q4. What is the concept of standardization (z-score), and why is it important in data analysis? Explain the formula and how standardization transforms a dataset

Answer - The concept of a Standard Score (or standardization) is used to convert data values into a common scale so they can be easily compared.

1. What is Standardization (Z-score)?

Standardization is the process of transforming a data value so that it shows how many standard deviations the value is away from the mean of the dataset.

In simple terms:

- It tells how far a value is from the average.
- The result is called a Z-score**.**

Example idea:

- If a value has Z = 2, it means the value is 2 standard deviations above the mean**.**
- If Z = −1, it means the value is 1 standard deviation below the mean**.**

2. Formula of Z-score

The formula is:

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z = (x − M) / Σ
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Where:

Symbol	Meaning
Z	Z-score (standardized value)
X	Original data value
M	Mean of the dataset
Σ	Standard deviation

3. How Standardization Transforms Data

When we standardize a dataset:

- The mean becomes 0
- The standard deviation becomes 1
- The data is transformed into a standard normal distribution scale

Example:

Original value (x)	Mean	Std Dev	Z-score
60	50	5	(60−50)/5 = **2**

Interpretation:
The value 60 is 2 standard deviations above the mean**.**

4. Why Standardization is Important in Data Analysis

Standardization is important because it helps:

1. Compare values from different datasets
Example: comparing exam scores from different subjects.

2. Identify outliers
Values with very high or very low z-scores may be unusual.

3. Prepare data for statistical models
Many algorithms work better when variables are standardized.

4. Understand relative position of data points
It shows whether a value is **above or below average and by how much**.

Q5. What is Kurtosis and their type?

Answer – Kurtosis is a statistical measure that describes the shape of a distribution**,** specifically how peaked or flat the data distribution is compared to a normal distribution and how heavy the tails are.

In simple terms, kurtosis tells us:

- How sharp or flat the peak of the distribution is.
- How extreme values (outliers) appear in the data.

Types of Kurtosis

There are three main types.

1. Mesokurtic Distribution

- Kurtosis value ≈ **3**
- Shape similar to a normal distribution
- Moderate peak and moderate tails.

Example: Many naturally occurring datasets follow this pattern.

2. Leptokurtic Distribution

- Kurtosis value greater than 3
- Very sharp peak
- Heavy tails (more extreme values or outliers)

Meaning: Data is more concentrated near the mean but has more extreme values**.**

3. Platykurtic Distribution

- Kurtosis value less than 3
- Flatter peak
- Light tails

Meaning: Data is more evenly spread out and has fewer extreme values.

Q6. Explain why the uniform distribution is a good model for the outcome of rolling a fair die

Answer – The uniform distribution is a good model for the outcome of rolling a fair die because each of the six possible outcomes (1–6) has an equal probability of occurring. Since all outcomes are equally likely and there is no bias toward any particular number, the probability distribution is uniform.

Q7. Use Excel to compute the probability of getting at least 8 successes in 15 trials with success probability 0.5

Answer –

Q8. How does log transformation help in stabilizing variance and making data more normally distributed?

Answer - Log transformation is a technique in which the logarithm of the data values is taken. It helps stabilize variance by compressing large values and reducing differences in variability. It also reduces skewness and makes the data distribution more symmetric, often bringing it closer to a normal distribution. This improves the reliability of statistical analysis and modelling.